

SUR, Oman

WMO: 412680

Lat: 22.533N

Lon: 59.483E

Elev: 46

StdP: 14.67

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.3	63.2	33.4	28.0	85.7	37.7	33.2	86.3	25.7	74.9	23.4	75.9	5.1	230	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	16.6	110.2	76.2	107.8	75.7	105.7	75.7	86.7	98.3	85.1	96.5	83.6	94.9	14.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.1	179.0	94.9	82.3	168.4	93.4	80.7	159.5	92.5	51.8	98.3	49.8	97.1	47.9	95.0	97.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.5	28.3	25.1	56.9	115.2	1.2	1.9	56.1	116.5	55.4	117.7	54.7	118.7	53.8	120.1		
			48.8	86.9	2.2	3.6	47.3	89.5	46.0	91.6	44.8	93.7	43.2	96.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	85.2	71.6	74.2	79.9	88.2	95.8	96.8	93.8	91.7	90.6	87.0	79.0	73.5
	DBStd	9.65	3.10	4.38	5.18	5.29	4.56	4.29	4.45	4.48	3.42	3.77	3.35	2.94
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12862	670	679	928	1145	1420	1403	1358	1293	1219	1147	870	730
	CDD65	7387	205	259	463	695	955	953	893	828	769	682	420	265
	CDH74	98205	726	1519	4425	9563	15398	15538	13907	12275	10858	8887	3765	1344
	CDH80	59919	76	377	1769	5544	10939	11219	9450	7854	6596	4766	1177	152
(14) Wind	WSAvg	10.9	7.7	9.4	9.8	10.5	13.7	14.1	14.3	13.7	12.9	9.5	7.2	7.4
(15) Precipitation	PrecAvg	3.7	0.5	0.9	0.6	0.4	0.0	0.2	0.1	0.1	0.0	0.1	0.3	0.4
	PrecMax	10.0	3.9	7.8	4.1	1.6	0.2	1.2	1.1	1.5	0.0	1.1	2.6	3.3
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.0	0.9	1.9	1.0	0.5	0.0	0.4	0.3	0.3	0.0	0.2	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.0	91.6	99.5	106.8	112.7	113.4	111.9	110.3	106.4	102.5	94.1	86.7
		MCWB	67.3	65.6	67.0	72.1	77.1	75.1	76.8	78.2	74.6	69.7	68.9	68.6
	2%	DB	81.9	87.3	95.1	102.9	109.3	110.3	108.0	105.6	103.0	99.2	90.7	83.4
		MCWB	67.0	66.3	66.3	71.0	74.4	76.3	77.0	77.3	75.5	71.4	69.1	68.6
	5%	DB	79.8	84.4	91.8	99.8	106.7	107.9	105.1	102.4	100.3	96.8	88.1	81.4
		MCWB	67.0	66.3	66.5	70.4	73.8	76.5	77.3	77.5	75.8	71.8	70.3	68.4
	10%	DB	77.8	81.7	88.5	96.9	104.3	105.4	102.1	99.7	97.8	94.4	85.9	79.8
		MCWB	66.5	66.2	66.6	69.7	73.6	77.1	78.1	77.9	75.7	71.9	71.3	68.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.8	74.4	76.9	82.4	88.9	88.9	87.7	87.6	86.9	84.3	79.2	75.0
		MCDB	78.6	80.9	83.9	98.8	104.3	101.8	99.1	96.2	95.8	91.5	85.8	80.2
	2%	WB	71.6	72.2	75.2	79.4	85.1	86.3	85.5	85.0	84.2	81.4	76.9	73.0
		MCDB	77.4	79.6	82.6	92.2	98.0	100.2	95.8	95.0	93.0	89.0	83.2	79.2
	5%	WB	69.9	70.7	73.7	77.4	82.7	84.7	83.7	82.9	82.1	79.4	75.3	71.3
		MCDB	76.2	78.5	82.1	89.8	95.4	97.3	94.5	94.7	92.1	88.3	82.4	78.1
	10%	WB	68.2	69.2	72.1	75.6	80.4	83.0	82.0	81.4	80.3	77.6	74.0	69.9
		MCDB	75.4	77.4	81.2	87.5	93.3	95.2	94.0	93.7	90.9	87.4	81.7	77.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.2	14.6	14.9	16.2	16.6	15.6	16.0	16.6	15.6	13.9	12.8
		MCDBR	15.2	17.1	18.8	18.2	18.7	20.3	20.4	19.6	19.4	17.9	16.5	14.7
		MCWBR	6.4	6.9	8.6	7.8	7.6	7.3	6.5	6.9	6.9	8.7	7.7	6.0
	5% WB	MCDBR	13.0	13.9	13.9	15.3	15.9	15.6	14.0	14.6	15.2	14.2	13.3	12.8
		MCWBR	7.0	7.3	8.4	8.3	8.7	8.3	8.9	9.1	9.1	9.6	7.7	6.3
(40) Clear-Sky Solar Irradiance	taub		0.428	0.471	0.536	0.602	0.653	0.740	0.815	0.750	0.621	0.487	0.451	0.427
	taud		2.175	2.029	1.854	1.741	1.654	1.522	1.447	1.532	1.717	1.988	2.111	2.173
	Ebn at Noon		263	258	247	233	220	200	186	198	222	250	252	257
	Edh at Noon		43	53	66	75	81	92	99	91	74	54	45	42
(44) All-Sky Solar Radiation	RadAvg		1407	1663	1914	2097	2238	2101	1938	1949	1902	1754	1472	1337
	RadStd		59	63	89	63	35	87	98	48	33	45	53	56

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.41	N/A	N/A	N/A	+0.70	+0.81	N/A	N/A	+173	+176	

Nomenclature: See separate page